

Evidence objects

D_{emp}
observed grouped curves

$D_{path}(\alpha)$
deterministic rule audit

D_{sim}
stochastic simulation
not reported

Declare context θ
 $\mathcal{M}, M_0$, groups
 n_{eff}, C

Fit candidates
SSE + BIC_{eff}

Grouped holdout
 $E_{CV} + XPE$

Evidence ledger $\mathcal{L}(D, \mathcal{M})$
 $C_\lambda(M) = BIC_{eff}(M) - \lambda_n \log[XPE(M)]$
 $\delta_\theta^{RIEC}[\mathcal{L}] = \arg \min_M C_\lambda(M)$

Auditable output

M_{RIEC}
 $G_\theta(M)$: score gaps
 G_θ^* : runner-up gap

Conditional
recommendation
+ tables + overlays

Pairwise switch B over A iff $\Delta BIC_{eff}(A, B) < \lambda_n \log[E_{CV}(A)/E_{CV}(B)]$

L0: grouped evaluation world
(no point-wise leakage)

L1: pre-specified conflict resolution
(decision map + margins)

L2: review / deployment output
(conditional, auditable)